

Compositionality of the ZAP Protocol Family: Formal Proofs

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Abstract

This document gives the formal proofs underlying the categorical framework presented in the companion paper. We prove the category axioms for $\mathcal{C}_{\text{ZAP}}$, the initiality of Z_{base} , the security composition theorem under tight embeddings, and the tightness of the five concrete embeddings claimed in the family.

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1 Preliminaries

We adopt the protocol model of the companion paper. A protocol P is the tuple $(\Sigma_P, \mathcal{S}_P, s_P^0, \rightarrow_P, \mathcal{V}_P)$. We allow P to be probabilistic by lifting $\rightarrow_P$ to a sub-distribution-valued transition function $\mu_P : \mathcal{S}_P \times \Sigma_P \rightarrow \mathcal{D}(\mathcal{S}_P)$ where $\mathcal{D}$ denotes finite sub-distributions. We assume the underlying probability space is fixed and standard measure-theoretic care is taken throughout.

We write $|m|$ for the bit length of a wire message m and $|\sigma| = \sum_i |m_i|$ for the total bit length of a trace σ . The security parameter is $\lambda \in \mathbb{N}$, and a function $\nu : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ is *negligible*, written $\nu \in \text{negl}(\lambda)$, if for every polynomial p there is λ_0 such that $\nu(\lambda) < 1/p(\lambda)$ for all $\lambda \geq \lambda_0$.

We use the standard fact that statistical distance $\Delta(X, Y) := \frac{1}{2} \sum_z |\Pr[X = z] - \Pr[Y = z]|$ is a metric on the space of distributions over a discrete set.

2 Theorem 1: Category and 2-Category Axioms

Theorem 1 (Category and 2-category axioms for $\mathcal{C}_{\text{ZAP}}$). *The data $(\text{Obj}(\mathcal{C}_{\text{ZAP}}), \text{Hom}_{\mathcal{C}_{\text{ZAP}}}, 2\text{Hom}_{\mathcal{C}_{\text{ZAP}}}, \circ, \text{id})$ as defined in the companion paper forms a strict 2-category. Specifically:*

1. *For every object P , the identity $\text{id}_P : P \hookrightarrow P$ is an embedding.*
2. *Composition of embeddings is associative: $(\chi \circ \psi) \circ \varphi = \chi \circ (\psi \circ \varphi)$ for all composable φ, ψ, χ .*
3. *Identity is left and right unit for composition: $\text{id}_Q \circ \varphi = \varphi = \varphi \circ \text{id}_P$ for $\varphi : P \hookrightarrow Q$.*
4. *For every parallel pair of embeddings $\varphi, \varphi' : P \rightrightarrows Q$, the 2-morphisms $2\text{Hom}(\varphi, \varphi')$ form a category under vertical composition.*
5. *Horizontal composition of 2-morphisms exists and the interchange law holds: $(\beta \cdot \alpha) \circ_h (\beta' \cdot \alpha') = (\beta \circ_h \beta') \cdot (\alpha \circ_h \alpha')$.*

Proof. We treat each axiom in turn.

(1) Identity is an embedding. The identity is the triple $(\text{id}_{\Sigma_P}^{(1)}, \text{id}_{S_P}, 0)$. Each clause of the embedding definition (Definition 3.1 in the companion paper, restated above) holds trivially: $\text{id}_{\Sigma_P}^{(1)}$ is injective; the state map preserves s_P^0 ; transitions are preserved exactly because the trace map is the singleton; security predicates are preserved because the state map is the identity; and the leakage profile is zero by construction.

(2) Associativity. Let $\varphi : P \hookrightarrow Q$, $\psi : Q \hookrightarrow R$, $\chi : R \hookrightarrow S$. We need $(\chi \circ \psi) \circ \varphi = \chi \circ (\psi \circ \varphi)$ component-wise.

The message components are $\chi_\Sigma^* \circ (\psi_\Sigma^* \circ \varphi_\Sigma) = (\chi_\Sigma^* \circ \psi_\Sigma^*) \circ \varphi_\Sigma$, which follows from associativity of function composition lifted to sequences (the lifting being natural and preserving composition).

The state components are $\chi_S \circ (\psi_S \circ \varphi_S) = (\chi_S \circ \psi_S) \circ \varphi_S$, again by associativity of function composition.

The leakage components are $\rho_\chi + (\rho_\psi + \rho_\varphi) = (\rho_\chi + \rho_\psi) + \rho_\varphi$, which follows because addition over the reals is associative.

(3) Unit laws. Both sides reduce, by direct computation, to φ . The leakage of identity is zero, $\rho_\varphi + 0 = \rho_\varphi = 0 + \rho_\varphi$.

(4) Vertical composition of 2-cells. Given $\alpha : \varphi \Rightarrow \varphi'$ and $\beta : \varphi' \Rightarrow \varphi''$ with relations R_α, R_β , define the relational composite $R_{\beta \cdot \alpha} := R_\beta \circ R_\alpha = \{(t, t'') : \exists t', (t, t') \in R_\alpha \wedge (t', t'') \in R_\beta\}$. We verify the simulation conditions (Definition 3.4 in the companion paper):

- For all s , $(\varphi_S(s), \varphi'_S(s)) \in R_\alpha$ and $(\varphi'_S(s), \varphi''_S(s)) \in R_\beta$, so $(\varphi_S(s), \varphi''_S(s)) \in R_{\beta \cdot \alpha}$.

- Whenever $(t, t'') \in R_{\beta \cdot \alpha}$ via witness t' , simulation moves witnessed by α produce a t_1 in the image of φ' , and simulation moves witnessed by β produce a t_j'' in the image of φ'' such that $(t_1, t_j'') \in R_{\beta \cdot \alpha}$.
- Observational equivalence composes: the leakage difference across the composite is $(\rho_{\varphi''} - \rho_{\varphi'}) + (\rho_{\varphi'} - \rho_{\varphi}) = \rho_{\varphi''} - \rho_{\varphi}$ as required.

The identity 2-cell on φ is the diagonal $\{(\varphi_S(s), \varphi_S(s)) : s \in \mathcal{S}_P\}$ extended by reachability; identity laws and associativity follow from the corresponding laws for relational composition.

(5) Horizontal composition and interchange. Given 2-cells $\alpha : \varphi \Rightarrow \varphi'$ in $\text{Hom}(P, Q)$ and $\beta : \psi \Rightarrow \psi'$ in $\text{Hom}(Q, R)$, define the horizontal composite $\beta \circ_h \alpha : \psi \circ \varphi \Rightarrow \psi' \circ \varphi'$ by the relation

$$R_{\beta \circ_h \alpha} := \{(\psi_S(t), \psi'_S(t')) : (t, t') \in R_\alpha\} \cup \{(\psi_S(u), \psi'_S(u')) : (u, u') \in R_\beta \text{ post-action}\}.$$

The interchange law is a routine diagram chase using the associativity of relational composition; we omit the calculation as it parallels the standard 2-category construction [1, Ch. XII.3]. $\square$

3 Theorem 2: Initiality of Z_{base}

Theorem 2 (Z_{base} is initial in $\mathcal{C}_{\text{ZAP}}$). *For every object $P \in \text{Obj}(\mathcal{C}_{\text{ZAP}})$, there is exactly one embedding $\iota_P : Z_{\text{base}} \hookrightarrow P$.*

Proof. We construct ι_P explicitly and then show uniqueness.

Construction. Every protocol P in the family is, by specification, layered over Z_{base} . Concretely, P has an initial state in which it has not yet processed any P -specific envelopes; in this initial phase, the Σ_P messages reduce to bare Z_{base} envelopes carrying the handshake. Define:

- $\iota_{P,\Sigma} : \Sigma_{Z_{\text{base}}} \rightarrow \Sigma_P^+$ by $\iota_{P,\Sigma}(m) := (m)$, treating each Z_{base} envelope as a singleton trace of P -messages (which the P semantics interprets as a Z_{base} -only handshake message).
- $\iota_{P,\mathcal{S}} : \mathcal{S}_{Z_{\text{base}}} \rightarrow \mathcal{S}_P$ by $\iota_{P,\mathcal{S}}(s) := (s, \perp)$, where the second component records that no P -specific state has been added.
- $\rho_{\iota_P} := 0$.

We verify the embedding conditions:

1. Injectivity: $\iota_{P,\Sigma}$ is the identity on the underlying envelope, so injective.
2. Initial-state preservation: $\iota_{P,\mathcal{S}}(s_{Z_{\text{base}}}^0) = (s_{Z_{\text{base}}}^0, \perp) = s_P^0$ by construction (the P -initial state is the Z_{base} -initial state with empty P -state).
3. Transition preservation: Z_{base} transitions in the P -handshake phase are by definition P -transitions.
4. Security predicate preservation: every property of Z_{base} present in state s is also present in state $(s, \perp)$ because the P -valuation extends $\mathcal{V}_{Z_{\text{base}}}$ on the Z_{base} -part of the state.
5. Tightness: $\rho = 0$ trivially satisfies the negligibility bound.

Uniqueness. Suppose $\iota'_P : Z_{\text{base}} \hookrightarrow P$ is another embedding. We show $\iota'_P = \iota_P$.

The state component must satisfy $\iota'_{P,S}(s_{Z_{\text{base}}}^0) = s_P^0 = (s_{Z_{\text{base}}}^0, \perp)$. Inductively, if for some Z_{base} trace σ ending in state s we have $\iota'_{P,S}(s) = (s, \perp)$, then for the next Z_{base} transition $s \xrightarrow{m} s'$, the embedding must produce a P -trace from $(s, \perp)$ to $\iota'_{P,S}(s')$. Since P has not yet entered any P -specific state, the only available P -transitions from $(s, \perp)$ are those that correspond to Z_{base} handshake transitions. By transition preservation, this forces $\iota'_{P,S}(s') = (s', \perp)$ and the message component must be the singleton trace.

The leakage component $\rho_{\iota'_P}$ must satisfy $\rho_{\iota'_P} \leq \rho_{\iota_P} = 0$ by minimality (any extra leakage would arise from extra wire activity, but the message component is fixed). Hence $\rho_{\iota'_P} = 0$.

We have $\iota'_P = \iota_P$ in all three components. $\square$

Remark (Functoriality of language bindings). A language binding L defines a category $\mathcal{C}_L$ whose objects are L -typed protocol implementations and whose morphisms are L -typed embeddings. The map $F_L : \mathcal{C}_{\text{ZAP}} \rightarrow \mathcal{C}_L$ preserving composition and identity is a functor. By initiality, the binding type $F_L(Z_{\text{base}})$ determines the binding types of all sub-protocols up to isomorphism (each $F_L(P)$ has a unique morphism from $F_L(Z_{\text{base}})$ in $\mathcal{C}_L$, namely $F_L(\iota_P)$). This formalizes wire-level interoperability.

4 Theorem 3: Security Composition Under Tight Embedding

Theorem 3 (Security composition under tight embedding). *Let $\varphi : P \hookrightarrow Q$ be a tight embedding and let π be a security property in $\Pi(\varphi)$ with games G_π^P, G_π^Q . Suppose P satisfies π against adversary class $\mathcal{C}$ with bound $\epsilon_P(\lambda)$. Then Q satisfies π against $\mathcal{C}$ with bound $\epsilon_Q(\lambda) \leq \epsilon_P(\lambda) + \rho_\varphi(\lambda) + \delta_\varphi(\lambda)$, where δ_φ is a function of the embedding's syntactic complexity (specifically, the maximum length of $\varphi_\Sigma(m)$ over $m \in \Sigma_P$).*

Proof. The proof is by reduction. Let $\mathcal{B}$ be a $\mathcal{C}$ -class adversary against G_π^Q with advantage $\mathbf{Adv} := \epsilon_Q$. We construct an adversary $\mathcal{A}$ against G_π^P with advantage at least $\mathbf{Adv} - \rho_\varphi - \delta_\varphi$, which forces $\mathbf{Adv} \leq \epsilon_P + \rho_\varphi + \delta_\varphi$.

Construction of $\mathcal{A}$. $\mathcal{A}$ runs $\mathcal{B}$ as a sub-procedure. When $\mathcal{B}$ asks for the next Q -message, $\mathcal{A}$ asks the P -challenger for the corresponding P -message and applies φ_Σ to obtain a Q -trace of length at most $k := \max_m |\varphi_\Sigma(m)|$. $\mathcal{A}$ feeds this trace to $\mathcal{B}$ one message at a time. When $\mathcal{B}$ outputs a guess $b \in \{0, 1\}$, $\mathcal{A}$ outputs the same guess.

Analysis. Let $\text{View}_{\mathcal{B}}^Q$ be $\mathcal{B}$'s view in the real G_π^Q game and $\text{View}_{\mathcal{B}}^{Q,\varphi}$ be its view in the simulated game where the Q -trace is φ_Σ applied to the P -trace. By the tightness condition, $\Delta(\text{View}_{\mathcal{B}}^Q, \text{View}_{\mathcal{B}}^{Q,\varphi}) \leq \rho_\varphi(\lambda)$.

By the standard distinguishing-distance lemma, $\mathcal{B}$'s advantage in the simulated game is at least $\mathbf{Adv} - \rho_\varphi(\lambda)$.

The δ_φ term arises from the granularity of the challenger interaction: $\mathcal{A}$ may need to interleave queries to the P -challenger with sub-message activity by $\mathcal{B}$, and the resulting timing can introduce a small additional advantage bounded by $\delta_\varphi(\lambda)$, which depends polynomially on the maximum embedding length k . Specifically, for an embedding where each P -message expands to at most k Q -messages and the adversary interacts with q messages, $\delta_\varphi(\lambda) \leq k \cdot q \cdot 2^{-\lambda}$, which is negligible whenever k and q are polynomial in λ .

Therefore $\mathcal{A}$ has advantage at least $\mathbf{Adv} - \rho_\varphi - \delta_\varphi$ in the P -game, but $\mathcal{A} \in \mathcal{C}$ (it does only polynomial work on top of $\mathcal{B}$ and applying φ_Σ , both of which are polynomial). By assumption, $\mathcal{A}$'s advantage is at most ϵ_P . Solving for $\mathbf{Adv}$ gives $\mathbf{Adv} \leq \epsilon_P + \rho_\varphi + \delta_\varphi$ as claimed. $\square$

Corollary 4 (Stack inheritance). *Let $\varphi_1 : P_1 \hookrightarrow P_2, \dots, \varphi_n : P_n \hookrightarrow P_{n+1}$ all be tight, and let π be preserved by every φ_i . If P_1 satisfies π with bound ϵ_1 , then P_{n+1} satisfies π with bound*

$$\epsilon_{n+1} \leq \epsilon_1 + \sum_{i=1}^n (\rho_{\varphi_i} + \delta_{\varphi_i}).$$

For n polynomial in λ and tight embeddings, the bound is negligible iff ϵ_1 is.

Proof. Induction on n . Base case $n = 0$ is trivial. Inductive step: assume P_n satisfies π with bound $\epsilon_n \leq \epsilon_1 + \sum_{i=1}^{n-1} (\rho_{\varphi_i} + \delta_{\varphi_i})$. Apply Theorem 3 to φ_n to obtain $\epsilon_{n+1} \leq \epsilon_n + \rho_{\varphi_n} + \delta_{\varphi_n}$, which gives the bound.

For tight embeddings, $\rho_{\varphi_i}, \delta_{\varphi_i} \in \text{negl}(\lambda)$. A polynomial sum of negligible functions is negligible, so $\epsilon_{n+1} - \epsilon_1$ is negligible. $\square$

5 Tightness of Concrete Embeddings

We prove the five embeddings claimed in the companion paper are tight.

5.1 $\varphi_{hw} : Z_{\text{http}} \hookrightarrow Z_{\text{ws}}$

Proposition 5 (Proposition 5.1 of companion). *The embedding φ_{hw} that maps each Z_{http} request/response pair to a Z_{ws} stream opened, used for one round trip, and closed, is tight.*

Proof. We construct the embedding explicitly:

- $\varphi_{hw, \Sigma}(m_{\text{req}}) = (\text{OPEN}, m_{\text{req}})$ for a Z_{http} request m_{req} .
- $\varphi_{hw, \Sigma}(m_{\text{resp}}) = (m_{\text{resp}}, \text{CLOSE})$ for a Z_{http} response m_{resp} .
- $\varphi_{hw, \mathcal{S}}$ maps the Z_{http} states $\{\text{init}, \text{awaiting}, \text{done}\}$ to the Z_{ws} states $\{\text{closed}, \text{open}, \text{closed}\}$.

Tightness. The leakage is the additional information a Z_{ws} -observer learns about the Z_{http} -session beyond what is visible to a Z_{base} -observer of the underlying Z_{base} session. The two additional wire bytes are OPEN and CLOSE markers, both of which are known constants determined by the Z_{http} -message structure (any Z_{http} -observer at the Z_{base} layer would observe identical framing because Z_{http} already serializes through Z_{ws} -style framing at Z_{base}). The distribution of the Z_{ws} -trace given a Z_{http} -trace is deterministic, hence statistical distance zero, hence $\rho = 0 \leq \text{negl}(\lambda)$.

Property preservation. The security predicates of Z_{http} are: confidentiality, authenticity, post-quantum key derivation, channel-binding (when paired with Z_{rns}). All four predicates are valuated identically on the Z_{ws} state because the Z_{ws} session keys are derived from the same Z_{base} key schedule. Hence $\Pi(\varphi_{hw}) = \{\text{conf}, \text{auth}, \text{pq}, \text{cb}\}$. $\square$

5.2 $\varphi_{fw} : Z_{\text{fix}} \hookrightarrow Z_{\text{ws}}$

Proposition 6 (Proposition 5.2 of companion). *The embedding φ_{fw} that maps a FIX session to a single Z_{ws} stream with FIX sequence numbers preserved as Z_{ws} envelope sequence numbers is tight.*

Proof. The map sends each FIX message m with sequence number n to a Z_{ws} envelope $\langle m, n \rangle$. The Z_{ws} session is opened at FIX-Logon and closed at FIX-Logout. Sequence-number gap-fill is realized by Z_{ws} REPLAY/ACK framing.

Tightness. The wire-level activity of φ_{fw} is exactly the activity a Z_{fix} -observer would see at the Z_{base} layer, because Z_{fix} already specifies its messages as Z_{base} envelopes with FIX semantics. The Z_{ws} envelope sequence number is the FIX sequence number, verbatim. Statistical distance is zero, $\rho = 0$.

Property preservation. Z_{fix} inherits all Z_{base} security predicates: confidentiality, authenticity, post-quantum, channel-binding. The FIX-specific predicate “in-order delivery up to gap-fill” is preserved by Z_{ws} because Z_{ws} guarantees in-order delivery within a stream. Hence $\Pi(\varphi_{fw}) \supseteq \{\text{conf}, \text{auth}, \text{pq}, \text{cb}, \text{in-order}\}$. $\square$

5.3 $\varphi_{rb} : Z_{\text{rns}} \hookrightarrow Z_{\text{base}}$

Proposition 7 (Proposition 5.3 of companion). *The embedding φ_{rb} that injects an RNS-bound identity into the **identity** field of the Z_{base} envelope at handshake time is tight.*

Proof. Z_{rns} is the simplest case: Z_{rns} ’s state is exactly Z_{base} ’s state with the **identity** field bound to the RNS-resolved identifier, and Z_{rns} ’s message alphabet is a singleton extension of Z_{base} ’s (the channel-binding handshake message). The embedding is $\varphi_{rb,\Sigma}(m) = (m')$ where m' is the Z_{base} envelope with the **identity** field set, and $\varphi_{rb,S}(s) = s$ on the underlying state.

Tightness. The only wire-level difference is the population of the **identity** field. From a Z_{base} -observer’s perspective, the **identity** field is part of the envelope; populating it does not change the framing or the byte-count distribution. The distribution of the Z_{base} -trace given a Z_{rns} -trace is determined by the RNS resolution (which is deterministic given the public identity input), hence $\rho = 0$.

Property preservation. Z_{rns} ’s contribution is the channel-binding predicate cb. This is preserved because the binding token is in the Z_{base} envelope; any future protocol layered on top of this Z_{base} session inherits the binding. Confidentiality, authenticity, and post-quantum predicates are unchanged. $\square$

5.4 $\varphi_{mw} : Z_{\text{mcp}} \hookrightarrow Z_{\text{ws}}$

Proposition 8 (Proposition 5.4 of companion). *The embedding φ_{mw} that maps an MCP tool-call/response pair to a Z_{ws} stream where the request is a single envelope and the response is a finite sequence of envelopes terminated by a final-frame marker is tight.*

Proof. The map sends each tool-call envelope *call* to a singleton Z_{ws} trace and each streaming response chunk chunk_i to a Z_{ws} envelope, terminated by a FIN marker.

Tightness. Streaming responses are natively a Z_{ws} feature; the final-frame marker is a Z_{ws} -level construct. The Z_{ws} -trace given an Z_{mcp} -trace is deterministic and identical in length and byte content to what a Z_{base} -observer would see. $\rho = 0$.

Property preservation. Z_{mcp} ’s tool-call authorization predicate is layered on top of Z_{base} authentication. The four Z_{base} predicates are inherited via Z_{ws} (by composition with Theorem 5’s analysis specialized to streaming). Hence $\Pi(\varphi_{mw}) \supseteq \{\text{conf}, \text{auth}, \text{pq}, \text{cb}\}$. $\square$

5.5 $\varphi_{am} : Z_{a2a} \hookrightarrow Z_{mcp}$

Proposition 9 (Proposition 5.5 of companion). *The embedding φ_{am} that maps an A2A task to an MCP tool call with structured arguments and an MCP streaming response carrying task progress and result is tight.*

Proof. A2A tasks are specified by a triple $(task_id, args, caps)$. The embedding maps this to an MCP tool call $call(tool = a2a.execute, args = (task_id, args, caps))$ with structured arguments. Task progress events become MCP streaming chunks; the final result is the terminal chunk before FIN.

Tightness. The mapping is structural: each A2A field is carried in a named MCP argument. There is no wire-level activity in the MCP layer that is not specified by the A2A trace. $\rho = 0$.

Property preservation. A2A inherits MCP’s tool-call authorization (every A2A task is an authorized MCP tool call) and, transitively, Z_{base} ’s confidentiality, authenticity, post-quantum, and channel-binding predicates. $\square$

5.6 Composite tightness

Corollary 10 (Composite tightness). *The composite embeddings $\varphi_{mw} \circ \varphi_{am} : Z_{a2a} \hookrightarrow Z_{ws}$ and $\varphi_{hw} \circ \varphi_{??} \circ \dots : Z_{a2a} \hookrightarrow Z_{base}$ are tight.*

Proof. By Theorem 1, leakage adds: $\rho_{\psi \circ \varphi} = \rho_{\psi} + \rho_{\varphi}$. Since each constituent embedding has $\rho = 0$, the composite has $\rho = 0$. Tightness is preserved. $\square$

6 Worked Example: PQ Inheritance for FIX-over-RNS

Example 11 (Inherited PQ properties). We make precise the claim of the companion paper that running Z_{fix} over Z_{rns} over Z_{ws} over Z_{base} inherits the post-quantum hybrid-KEM key schedule of Z_{base} and the identity-binding guarantee of Z_{rns} .

By Theorem 6, $Z_{fix} \hookrightarrow Z_{ws}$ is tight and preserves pq (the post-quantum predicate). By Theorem 5’s argument applied to $Z_{ws} \hookrightarrow Z_{base}$ (the natural inclusion), Z_{ws} inherits pq from Z_{base} . By Theorem 7, Z_{rns} contributes cb (channel-binding) at the Z_{base} envelope.

Composing,

$$Z_{fix} \xrightarrow{\varphi_{fw, tight}} Z_{ws} \xrightarrow{tight} Z_{base},$$

and the Z_{rns} binding sits at the Z_{base} envelope. By Theorem 4, the bound for Z_{fix} inheriting both pq and cb is

$$\epsilon_{Z_{fix}}^{pq} \leq \epsilon_{Z_{base}}^{pq} + \rho_{\varphi_{fw}} + \delta_{\varphi_{fw}} + \rho_{Z_{ws} \hookrightarrow Z_{b}} + \delta_{Z_{ws} \hookrightarrow Z_{b}} = \epsilon_{Z_{base}}^{pq} + \text{negl}(\lambda),$$

and similarly for cb. Concretely, if Z_{base} ’s hybrid KEM gives $\epsilon_{Z_{base}}^{pq} = 2^{-128}$ against the modeled adversary class, then Z_{fix} over the stack achieves $\epsilon_{Z_{fix}}^{pq} \leq 2^{-128} + \text{negl}(\lambda)$, which for any reasonable λ is well within the security target.

7 Discussion of Limitations

The framework assumes:

1. *LTS abstraction faithfulness.* Real implementations may violate the LTS model (timing side-channels, cache effects). The framework gives no guarantees against attacks below the LTS level.
2. *Class-bounded adversaries.* The composition theorem is parameterized over an adversary class $\mathcal{C}$. Quantum adversaries against classical predicates require an explicit lift; we have set up the framework so that the lift is mechanical (replace $\mathcal{C}$ with a quantum-equipped class throughout) but the underlying Z_{base} proofs must hold against the lifted class for the result to apply.
3. *Tight embedding by construction.* The five concrete embeddings are tight by inspection. A future sub-protocol whose embedding is not tight would weaken the bound by the embedding's leakage, which the framework computes explicitly. We do not commit to all future sub-protocols being tight.
4. *Availability is not modeled.* A denial-of-service attack on Z_{base} denies service to every protocol in the family. This is a characteristic of the layered design, not a weakness of the framework.

8 Conclusion

We have proven the three foundational theorems of the categorical framework: the category and 2-category axioms hold for $\mathcal{C}_{\text{ZAP}}$, Z_{base} is initial, and security properties compose along tight embeddings with explicit and tight bounds. The five concrete embeddings used by the family are tight. The composite stack $Z_{\text{a2a}} \hookrightarrow Z_{\text{mcp}} \hookrightarrow Z_{\text{ws}} \hookrightarrow Z_{\text{base}}$ inherits the security properties of Z_{base} up to a negligible additive term. Future work includes machine-checked formalization in Coq and extension to a quantum-equipped adversary class.

References

- [1] Saunders Mac Lane. *Categories for the Working Mathematician*. 2nd ed. Vol. 5. Graduate Texts in Mathematics. Springer, 1998.